DHANUSH DABBIRU

Senior Generative AI Engineer | **LLM Orchestration • Multimodal AI • RAG • Voice AI • Computer Vision**

703-626-3800 | kasiviswanadhdhanushdabbiru@gmail.com | GitHub: dkvd648

Canary | **Clarity Sidekick** | **Savoas** | **AfterBuy**

SUMMARY

Senior Generative AI Engineer building production **LLM tool-calling systems**, **RAG and persistent memory**, **multimodal video AI**, real-time **voice AI**, and cloud-native applications across **web**, **iOS**, **Android**, and **Smart TV**. Experienced across AI architecture, model orchestration, vector retrieval, backend APIs, computer vision, real-time speech, and production AWS deployment.

TECHNICAL SKILLS

AI / LLM: **LangGraph**, **LangChain**, **OpenAI Agents SDK**, **AWS Bedrock**, **Claude**, **Gemini**, **OpenAI**, **LLaMA 4 Scout**, **Groq**, **Amazon Nova**

RAG / Vision / Voice: **Pinecone**, **pgvector**, **DynamoDB**, **PostgreSQL**, **Redis**, **Multimodal RAG**, **OpenCV**, **Motion Detection**, **YOLOv12n**, **YOLOE**, **InsightFace**, **Deepgram Nova-3**, **Cartesia Sonic-3.5**, **Twilio / Telnyx**

Engineering / Cloud: **Python**, **C/C++**, **TypeScript**, **FastAPI**, **Flask**, **WebSockets**, SQLAlchemy, **AWS EC2 / ECS / ECR / S3 / Lambda**, **Docker**, **Terraform**, **Git / GitHub**, **React**, **Flutter**, **iOS**, **Android**, **Samsung Tizen**, **LG webOS**

PROFESSIONAL EXPERIENCE

AreaProbe Inc. | *Washington, DC* **Sep 2024 – Present**

Foundational Senior Generative AI Engineer

- Architected **SpeakCanary**, a multimodal video-intelligence platform processing **8–12 concurrent camera streams** with near-real-time motion, person/custom-object and facial-recognition detection, generating alerts in approximately **2–3 seconds**.
- Built the computer-vision pipeline using **OpenCV motion detection**, **YOLOv12n** for customer/person detection, **YOLOE** for custom detection, and **InsightFace** for facial recognition.
- Built **multimodal RAG** using **S3** frame storage, **Amazon Nova multimodal embeddings**, **Pinecone**, **AWS Bedrock**, **LLaMA 4 Scout**, **LangGraph**, and **FastAPI**, indexing and semantically searching **10K+ video frames/images** with retrieval in **under 1 second**; deployed across **EC2/ECS** and edge workers.
- Shipped web, **iOS/Android**, and **Samsung Tizen / LG webOS TV** applications. Also built **Ribbon** AI services and **Ribbon Images iOS**, plus **Anacostia** using **Flask**, MySQL, SQLAlchemy, and OpenAI.

ClarityPath AI | *Remote* **Apr 2026 – Present**

Senior Generative AI Engineer

- Building **Clarity Sidekick** with **LangGraph** model/tool loops, **AWS Bedrock**, and **Gemini**, supporting parallel tool execution, streamed responses, knowledge retrieval, and memory-aware conversations.
- Designed an **11-type durable-memory architecture** with **9-source per-turn context assembly** across **DynamoDB**, **PostgreSQL**, **Pinecone + S3**, and **Redis**; rolling summaries and memory caching reduced repeated context/token usage by approximately **30–40%**, with memory writes protected by a **3-tier deduplication pipeline**.
- Built real-time voice services using **Deepgram Nova-3**, **Cartesia Sonic-3.5**, **WebSockets/SSE**, and **FastAPI**, delivering approximately **1.5–2.5 second** end-to-end conversational response latency, with **React/TypeScript** and **Flutter iOS/Android** clients.

AfterBuy | *Remote* **2026 – Present**

AI / Full-Stack Engineer

- Building **AfterBuy**, an AI receipt-management platform that converts receipt photos into searchable purchase records and spending insights, including monthly spend, merchant activity, returns, warranties, and purchase history.
- Built receipt understanding and Q&A using **AWS Bedrock + LLaMA 4 Scout**, **FastAPI**, and AWS services, processing **1K+ receipt images** with approximately **90–95% extraction accuracy** for common receipt fields; shipped the mobile application with **Flutter** for **iOS** and **Android**.

xDial | *Remote* **Aug 2026 – Present**

AI Engineer

- Building healthcare voice/IVR infrastructure using **C**, **WebSockets**, **WebAssembly**, **NVIDIA Parakeet / Nemotron**, **Twilio**, **Medplum**, and PostgreSQL.
- Developed AWS infrastructure using **EC2 GPU workers**, **S3**, KMS, SSM, private networking, and **Terraform** for secure speech and call-processing workloads.

Quarterlife | *Remote* **Aug 2026 – Present**

AI Engineer

- Building an AI-assisted financial platform using **Claude**, **LangGraph**, structured tool calls, **Plaid**, **FastAPI**, and **PostgreSQL/Supabase**.
- Developed financial scoring and scenario-testing engines plus **React/TypeScript** and **Flutter iOS/Android** applications with safeguards for AI-generated financial outputs.

Kairos | *Remote* **Jun 2025 – Jan 2026**
Foundational AI Engineer
- Built a real-time AI voice assistant using **Deepgram STT/TTS**, **WebSockets**, **xAI Grok**, **Python**, and **Flask**, with Google Calendar, SendGrid, and SerpAPI integrations.

INDEPENDENT AI PROJECT

Savoas – AI Voice Agent – Built a real-time salon voice assistant using **LangGraph** model/tool loops, **Groq GPT-OSS**, **FastAPI**, **Twilio/Telnyx**, **Pinecone**, and **Redis**; integrated **20+ booking and business-action tools** for service lookup, availability, appointments, calendar operations, payments, and customer Q&A.

EDUCATION & CREDENTIALS

M.S. Computer Science, George Mason University – Machine Learning Concentration, GPA **3.47/4.00** 2023–2025
B.S. Computer Science, GITAM University – CGPA **8.44/10** 2019–2023

Certifications:
AWS Certified AI Practitioner (AIF-C02) • **AWS Certified Cloud Practitioner (CLF-C02)** • **Microsoft Azure AI-900**
Research:
"IoT Enabled Aqua Pond Monitoring System" (IEEE) • **"Semi-Automated Vehicle Control Using Wi-Fi" (Springer)**